

ECOTEC_Online Student Risk Dataset

Codebook and Metadata Documentation

This codebook documents the final 18-variable public release, the operational definition and provenance of TARGET_RISK, variable-level anonymization and generalization decisions, and the analytical scope of the dataset.

Dataset Overview

Attribute	Final public release
Institution	Universidad ECOTEC, Ecuador
Educational modality	Online higher education
Records	12,632 unique students
Variables	18
Null cells	0
Duplicate rows	0
Unique PUBLIC_RECORD_ID	12,632
Restricted institutional extraction	12,632 records x 42 variables

Target Distribution

TARGET_RISK	n	%
0	691	5.47
1	11,941	94.53

Interpretive limitation. TARGET_RISK is a contemporaneous administrative proxy constructed from the same institutional snapshot as the released predictors. It is not a prospectively observed dropout or retention outcome.

Variable Dictionary - Part I

Variable	Description	Type	Missing	Unique
PUBLIC_RECORD_ID	Release-specific anonymous record identifier replacing the restricted institutional student-record code.	Numeric	0	12632
FACULTY	Academic faculty associated with the released student record.	Categorical/Text	0	1
DEGREE_PROGRAM	Academic degree program represented as a standardized public category.	Categorical/Text	0	29
CURRICULUM_YEAR	Curriculum or study-plan year associated with the degree program.	Numeric	0	4
GENDER	Student gender represented as a standardized categorical variable.	Categorical/Text	0	2
AGE	Student age in completed years. Integer age is retained and is not generalized into age bands.	Numeric	0	51
MARITAL_STATUS	Student marital-status category standardized for public dissemination.	Categorical/Text	0	6
ETHNICITY	Student ethnicity represented through standardized public categories.	Categorical/Text	0	6
DISABILITY	Binary disability-status indicator. Detailed disability type and percentage are not included in the public release.	Categorical/Text	0	2

Variable Dictionary - Part II

Variable	Description	Type	Missing	Unique
INSTITUTION_TYPE	Generic institution-type category. Institution-specific names are not included in the public release.	Categorical/Text	0	5
NATIONALITY_COUNTRY	Generalized nationality category. Country-specific nationality values were aggregated into Ecuadorian and International.	Categorical/Text	0	2
TRANSFER_STUDENT_FLAG	Binary indicator identifying verified transfer-student status.	Categorical/Text	0	2
PAYMENT_PLAN	Non-monetary institutional payment-plan category.	Categorical/Text	0	2
ONLINE_ADMISSION_TYPE	Standardized category describing the online admission pathway.	Categorical/Text	0	15
PAYMENT_DEFERRAL_TYPE	Non-monetary institutional payment-deferral category.	Numeric	0	25
SCHOLARSHIP_NAME	Standardized scholarship or discount category. Monetary scholarship amounts are not included.	Categorical/Text	0	9
TARGET_RISK	Binary contemporaneous administrative target derived during restricted institutional curation from STUDENT_TYPE and GRADUATION_DATE. Class 0 indicates alumni status or a valid graduation date; class 1 indicates that the student was not recorded as alumni and no valid graduation date was available. STUDENT_TYPE and GRADUATION_DATE are excluded from the public predictor matrix.	Numeric	0	2
REGION	Generalized geographic region derived after restricted institutional reconciliation of province, canton, and, when required, verified sector information. Public categories are Coastal Region, Highlands Region, Amazon Region, and Insular Region.	Categorical/Text	0	4

TARGET_RISK Operational Definition

TARGET_RISK = 0: STUDENT_TYPE was Alumni or a valid GRADUATION_DATE was recorded.

TARGET_RISK = 1: the student was not recorded as Alumni and no valid GRADUATION_DATE was recorded.

No grade threshold, academic-performance cutoff, statistical model, weighted risk score, or institutional risk index was used to define TARGET_RISK. STUDENT_TYPE and GRADUATION_DATE were excluded from the public dataset and from the benchmark predictor matrix after target construction.

Privacy-Preserving Transformations and Public Generalization

Variable / field	Public treatment	Rationale
Institutional student-record code	Replaced one-to-one with release-specific PUBLIC_RECORD_ID.	Preserves within-release record linkage without exposing the institutional record code.
AGE	Retained as integer age; not converted to age bands.	Preserves analytical utility while direct personal identifiers are absent from the public dataset.
DISABILITY	Only binary Yes/No status retained; detailed disability type and percentage excluded.	Reduces sensitive detail while retaining accessibility-related analytical utility.
NATIONALITY_COUNTRY	Country-specific values generalized to Ecuadorian and International.	Reduces disclosure risk from rare country-specific categories while preserving domestic/international analysis.
REGION	Detailed geography reconciled internally and generalized to Coastal Region, Highlands Region, Amazon Region, and Insular Region.	Removes province-, canton-, and sector-level granularity while retaining broad geographic utility.
Institutional / financial detail	Institution-specific names, higher-granularity administrative tracking variables, and monetary amounts are not released.	Data minimization while retaining analytically useful non-monetary categories.

The restricted research extract used for this release did not include names, national identification numbers, email addresses, or telephone numbers. The institutional student-record identifier present in the restricted data was replaced during anonymization.

Final Generalized Public Categories

Variable	Category	n
NATIONALITY_COUNTRY	Ecuadorian	12,542
NATIONALITY_COUNTRY	International	90
REGION	Coastal Region	11,063
REGION	Highlands Region	1,338
REGION	Amazon Region	190
REGION	Insular Region	41

The public preprocessing workflow acts on the already anonymized 18-variable dataset and performs no additional anonymization.